

flyPad descent calculator

Top of descent, time, distance and fuel of the descent with the FMS performance model, and a live check of where to start down: the IN-FLT PERF DES module of the A380 OIS on both aircraft

Aircraft	A380X and A32NX
Where to find it	flyPad > Performance > Top of Descent tab
PR branch	pr/24-efb-descent-calculator (stacked on PR 23)
Status	Built, deployed and checked in the local flyPad harness; the in-sim test is still to do.
Pictures	flyPad pages in the local browser harness, running the build installed in the sim (A380X). The harness has no FMS, so the inputs are typed; in the sim, Fill data from FMS fills them.

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1. What it is for

The FMS computes the top of descent (T/D) for the managed descent. The crew still needs other answers: how long an emergency descent takes, where to start a descent at a given V/S, whether the aircraft is late on the profile after a shortcut, and a cross-check of the FMS T/D. The A380 OIS has an in-flight descent module for this (FCOM PER-IFT-DES); this page does the same on both aircraft, and replaces the old Top of Descent calculator.

It uses the **same flight and engine models as the FBW FMS predictions**, so its T/D agrees with the one drawn on the ND for the same inputs.

2. The screen

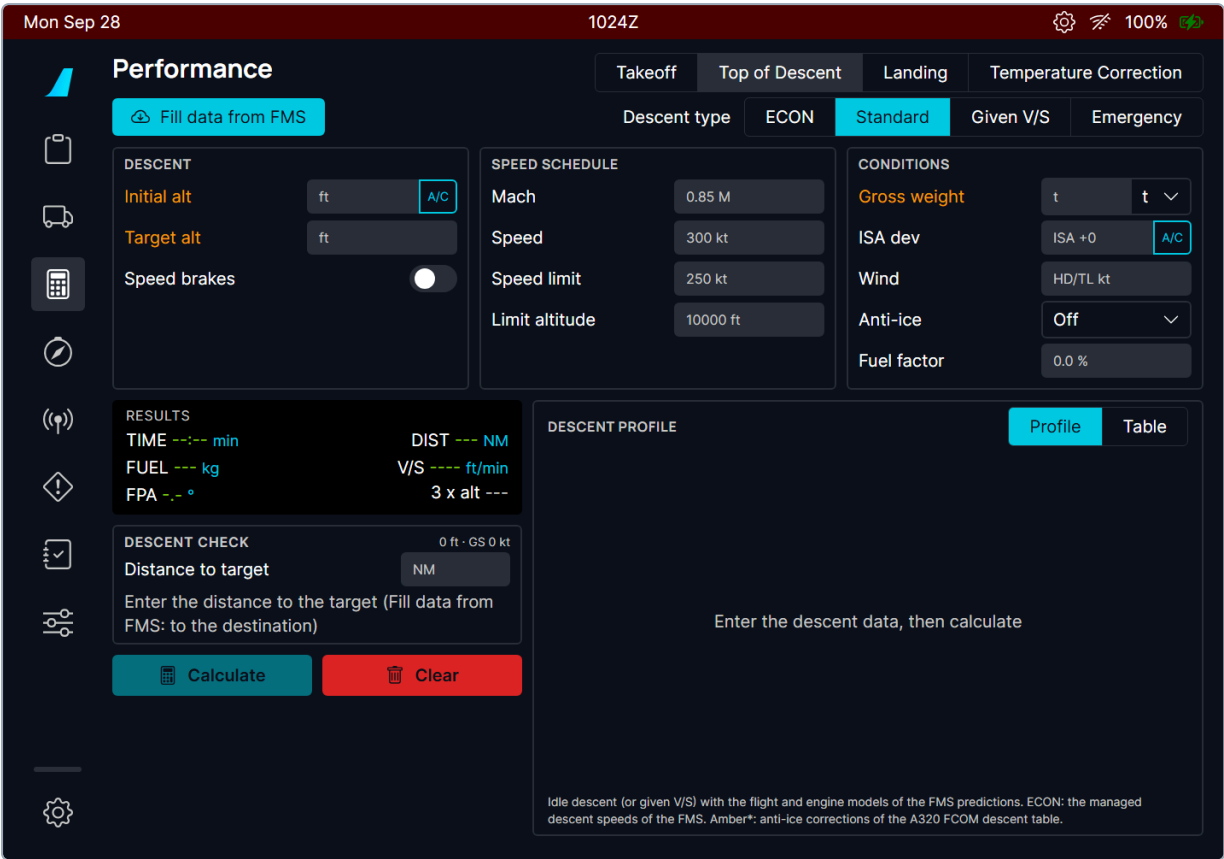


Figure 1. The descent calculator. Amber labels: inputs still needed.

Part	Contents
Fill data from FMS	Cruise level (or the current altitude when lower), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
Descent type	ECON : the managed descent speeds of the FMS. Standard : the FCOM speed schedule (A380 M.85 / 300 kt / 250 kt below FL100; A320 M.78 / 300 / 250). Given V/S : a constant vertical speed. Emergency : MMO / VMO with the speed brakes.
DESCENT	Initial altitude (A/C : the current one) and target altitude; speed brakes.
SPEED SCHEDULE	Mach, speed, speed limit and its altitude. Empty = the standard values shown in grey.
CONDITIONS	Gross weight, ISA deviation (A/C : from the current temperature), wind component (positive = headwind, negative = tailwind), anti-ice (off / engine / total), fuel factor.
RESULTS	Time, distance, fuel, mean V/S and flight path angle, and the "3 × altitude" rule of thumb for comparison; the crossover altitude and where the aircraft slows to the speed limit.
DESCENT CHECK	With the distance to the target: when to start down, or how late you are.
DESCENT PROFILE	Profile chart or table every 5000 ft.

3. Standard descent

- 1. Press **Fill data from FMS** in cruise with a flight plan, or type the initial altitude, the target altitude and the weight.
- 2. Choose the descent type and, if needed, the wind component and anti-ice.

3. Press **Calculate**.

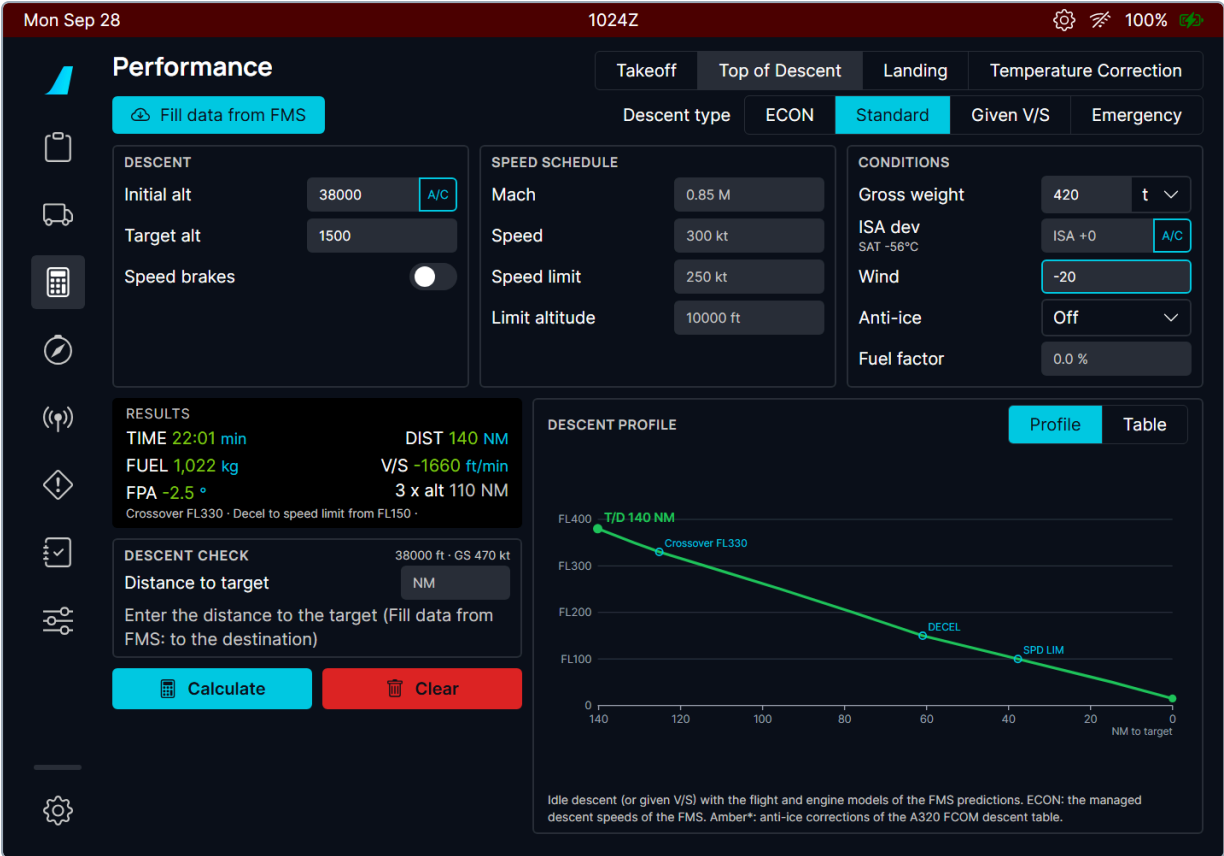


Figure 2. FL380 to 1500 ft, 420 t, 20 kt tailwind, standard speeds: 22 min, 140 NM, 1022 kg, -1660 ft/min on average (-2.5°). The chart shows the T/D, the crossover altitude (Mach to CAS), the deceleration (DECEL) and the speed limit (SPD LIM).

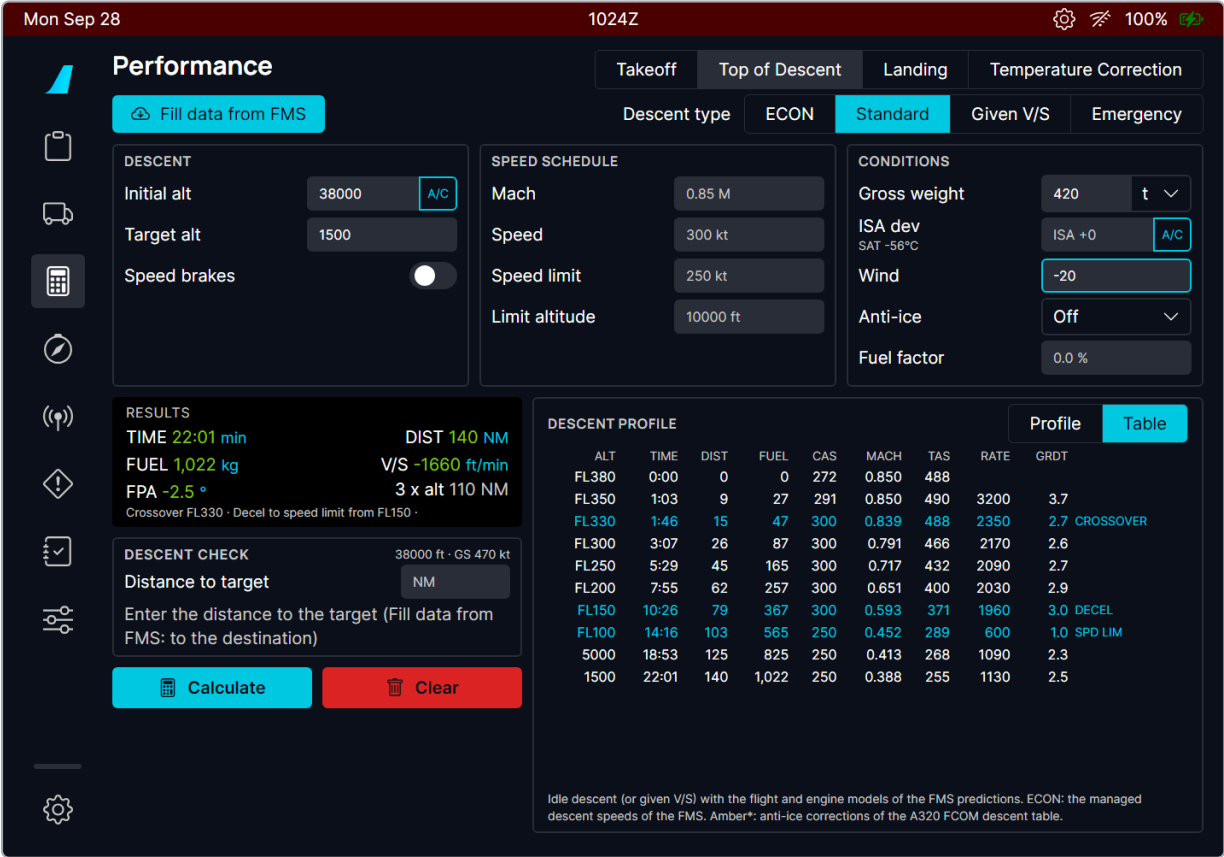


Figure 3. Table view: time, distance, fuel, CAS, Mach, TAS, RATE (rate of descent, ft/min) and GRDT (descent gradient, degrees) every 5000 ft, named and signed like the FCOM table results (PER-IFT-DES-DSR: both positive in descent); the crossover, DECEL and SPD LIM lines in blue. The RESULTS line gives V/S and FPA negative, as set on the FCU.

The deceleration to the speed limit is flown on the -1° idle path of the FMS descent strategy, which is why the profile flattens between DECEL and SPD LIM.

4. Descent check

Enter the **distance to the target** (Fill data from FMS enters the distance to the destination). The page reads the aircraft altitude and ground speed (shown top right of the box) and compares the aircraft with the profile. The aircraft is the triangle on the chart.

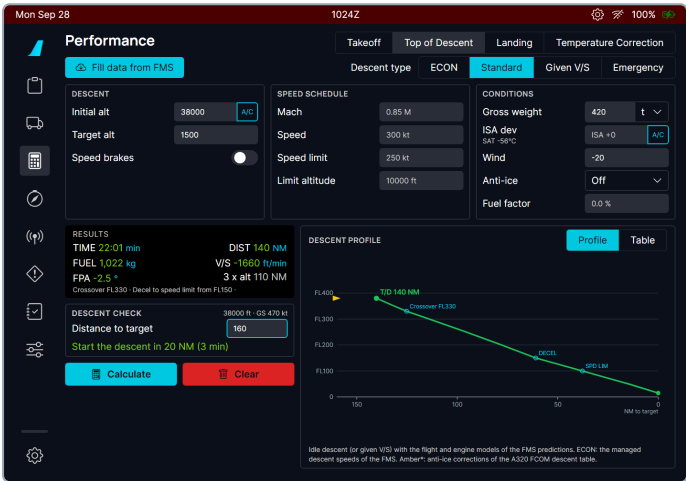


Figure 4. 160 NM to go, T/D at 140 NM: "Start the descent in 20 NM (3 min)".

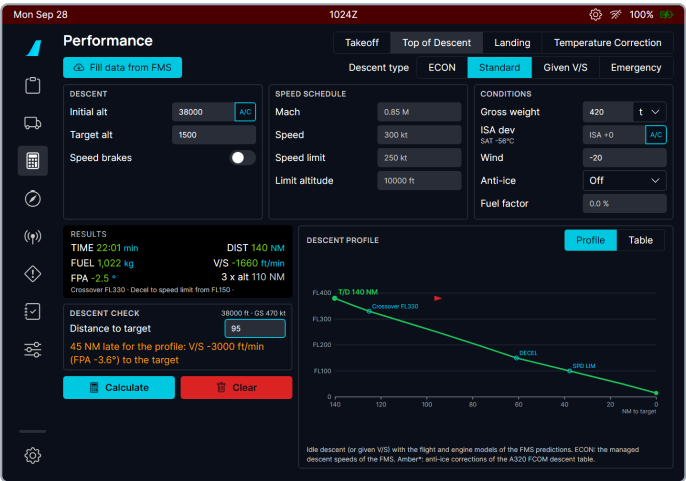


Figure 5. 95 NM to go at FL380: "45 NM late for the profile: V/S -3000 ft/min (FPA -3.6°) to the target".

TIP

After a shortcut, the late message gives the V/S (or FPA) that still makes the target, to set in V/S or FPA mode, or tells you to use the speed brakes.

5. Given V/S, emergency and ECON

Given V/S adds a V/S field. The page flies that V/S with the thrust it needs. Where the V/S is steeper than an idle descent can give, the page flies idle there and says so in amber (the aircraft cannot descend faster at idle without speed brakes).

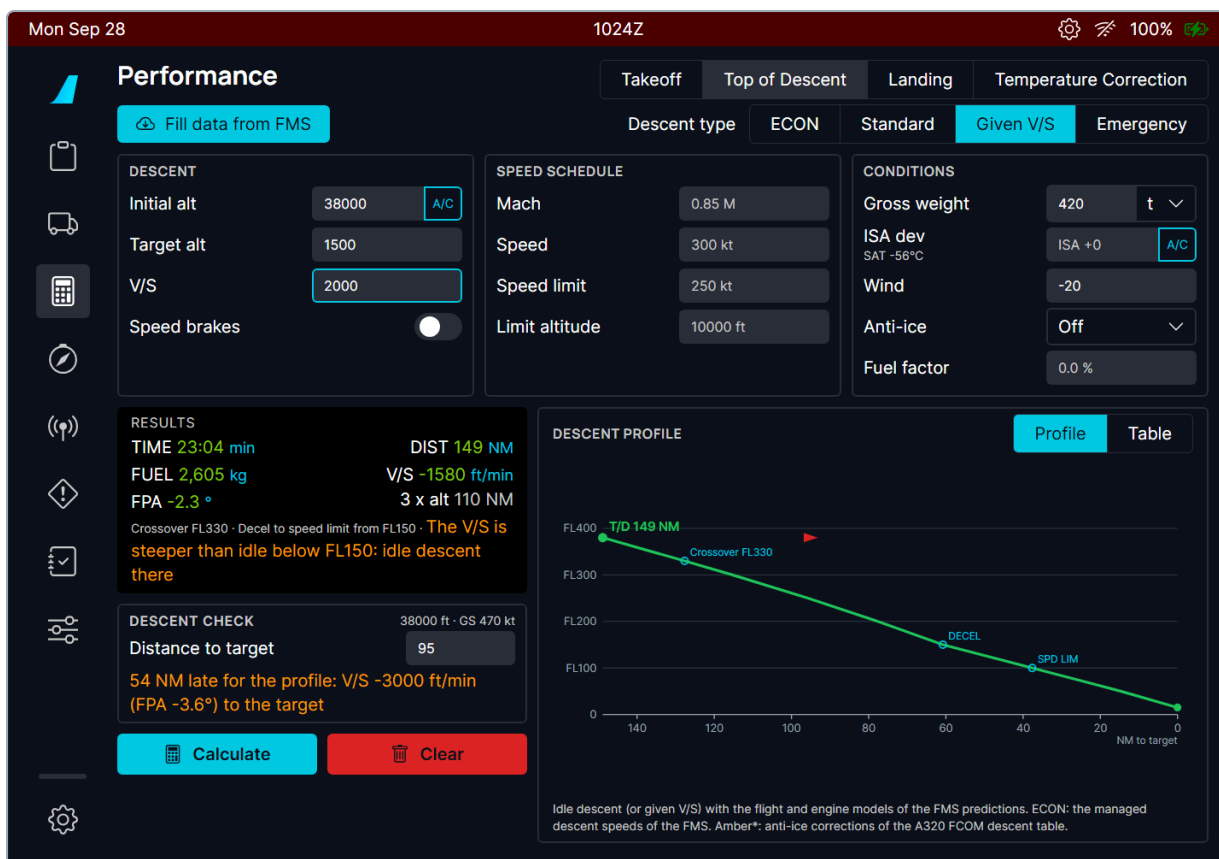


Figure 6. 2000 ft/min: shallower than idle at high level (149 NM, more fuel: 2605 kg), and steeper than idle below FL150, where the page flies idle.

Emergency uses MMO / VMO with the speed brakes extended (the schedule fields turn grey).

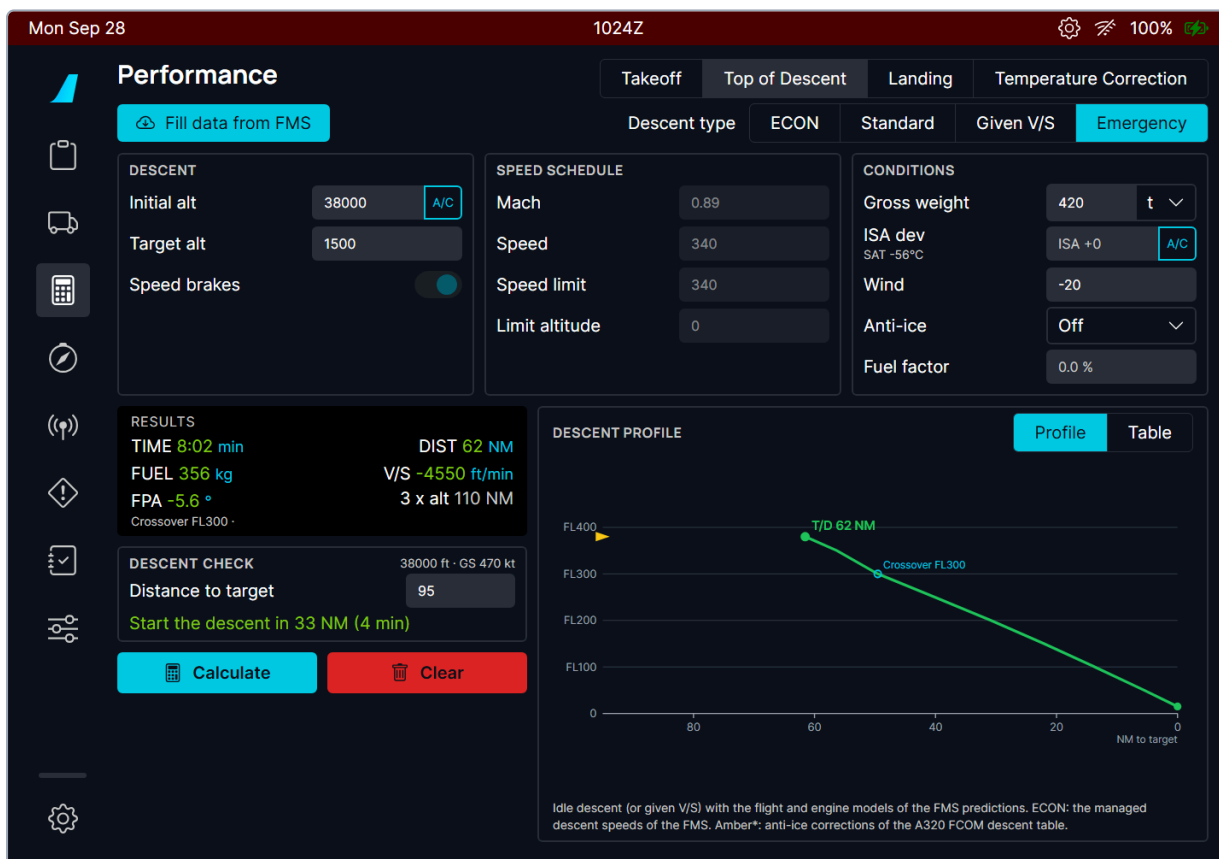


Figure 7. Emergency descent from FL380: 8 min, 62 NM, -4550 ft/min on average.

ECON needs the FMS: it asks the FMS for its managed descent speeds. Without a flight plan in the FMS, a message says so and the page stays on the previous type.

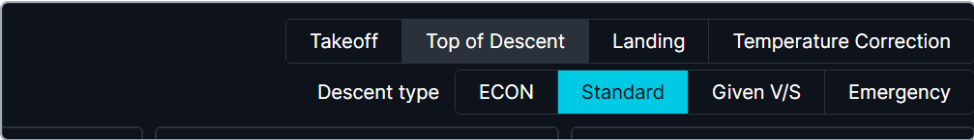


Figure 8. ECON without FMS data.

6. Anti-ice, fuel factor and estimates

Anti-ice (engine or total) raises the idle thrust and flattens the descent. Its effect uses the corrections of the A320 FCOM descent table (no A380 table is public): the values it changes turn **amber with a star**. The fuel factor (a percentage) scales the fuel burn, like the FMS PERF FACTOR.

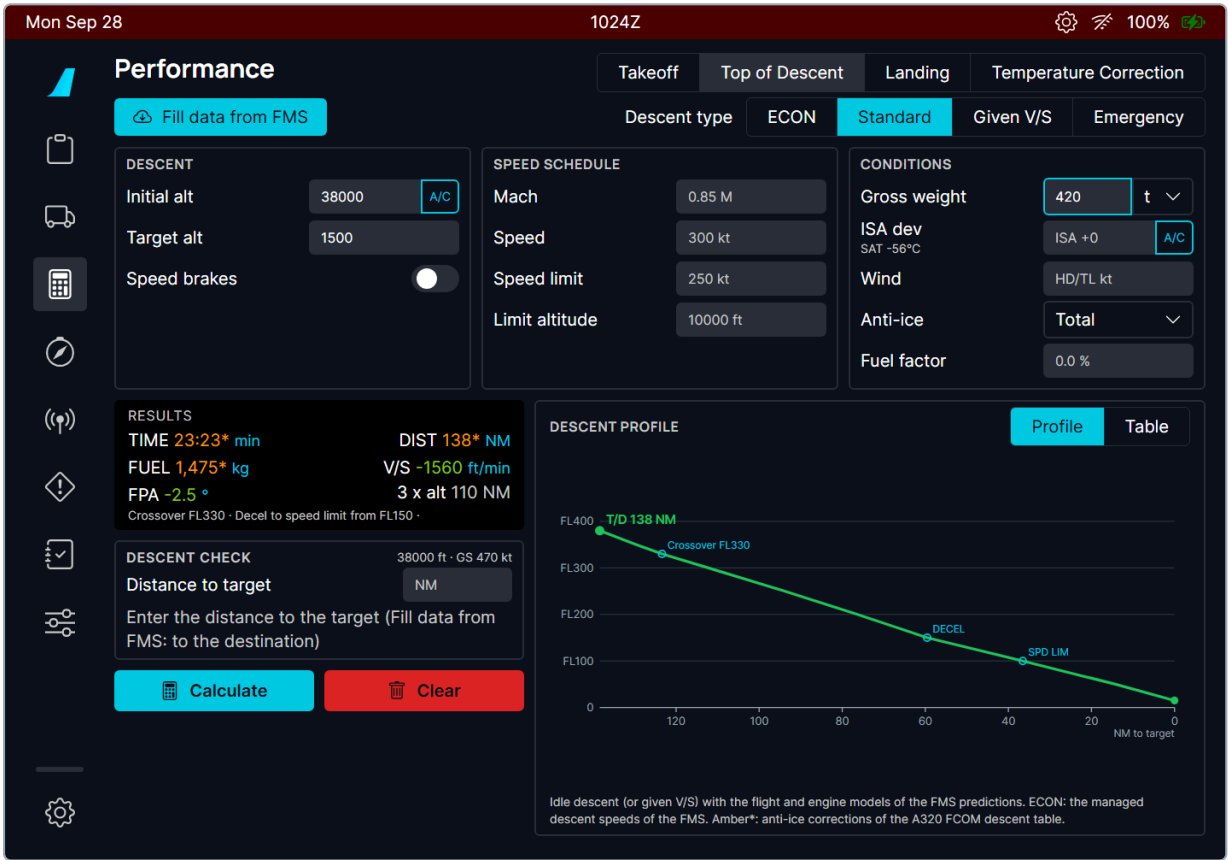


Figure 9. Total anti-ice: 138 NM, 23 min, 1475 kg, in amber (estimate).

7. What if the descent starts earlier or later?

After a calculation, the **T/D distance** slider under the profile moves the start of the descent: to the left farther from the target (earlier, e.g. vectors), to the right closer (later, e.g. a shortcut), up to 60 NM each way. The chart draws the moved top of descent dashed, straight to the target, and the line under the slider gives the V/S and the path angle it needs, to compare with the calculated ones.

- **Later:** steeper than the idle descent: the figures turn amber, "speed brakes or a higher speed needed" (red when the calculation already used the speed brakes: start earlier).
- **Earlier:** shallower than the idle descent: "thrust needed, more fuel".
- The figures keep the mean ground speed of the calculated descent: an estimate for the decision, not a new calculation.

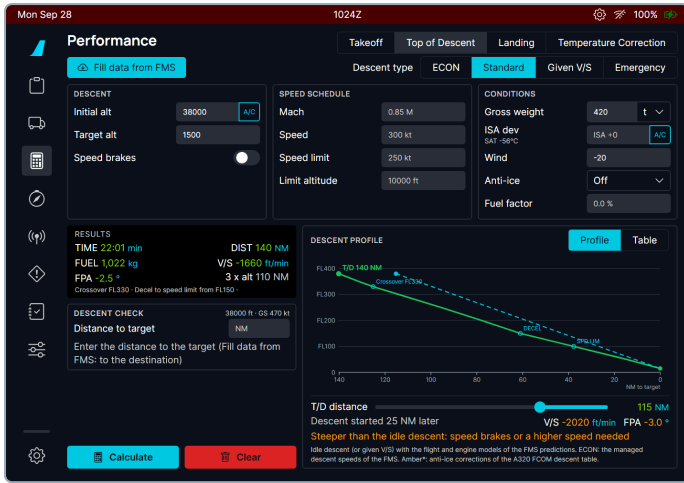


Figure 10. Slider 25 NM later: 115 NM, -2020 ft/min, -3.0°: steeper than idle.

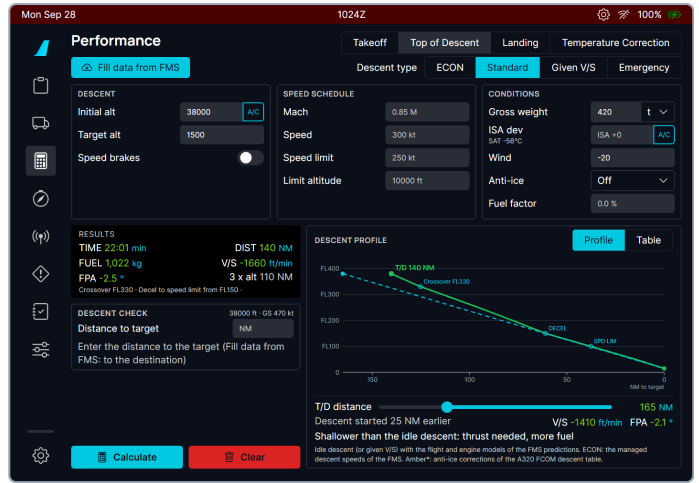


Figure 11. Slider 25 NM earlier: 165 NM, -1410 ft/min, -2.1°: shallower.

The **Calculate** and **Clear** buttons are at the bottom of the page, where they are on the takeoff and landing pages.

8. What to expect

Situation	What you should see
Inputs missing	Amber labels; no results.
Any input changed	The results are cleared: press Calculate again.
Target above the initial altitude	Message "The target altitude must be below the initial altitude".
Initial altitude above the aircraft ceiling, or weight out of range	A message gives the limit (maximum altitude, or the weight range from the empty weight to the MTOW).
Given V/S steeper than idle	Amber line: idle descent below the altitude it names.
ECON without FMS flight plan	"No FMS data: enter a flight plan in the FMS".
Distance to target larger than the descent	Green "Start the descent in X NM (Y min)".
Distance shorter than the descent	Amber "N NM late for the profile" with the V/S and FPA needed.

9. Limitations and why

Limitation	Why	What to do
The descent is a still-air profile plus a mean wind.	The wind is one component for the whole descent, as in the FCOM and OIS; the FMS uses the wind of each altitude.	Use the mean wind of the descent.
About 8 % longer than the A320 FCOM table.	The FCOM table is for the CFM56 A320; the FBW A320neo engine model idles differently.	Compare with the FMS T/D, which uses the same model.
Anti-ice values are estimates.	Only the A320 FCOM has anti-ice descent corrections.	-
No descent path constraints or approach.	The page computes a clean descent to one target altitude, not the FMS path with its constraints.	Use it as a cross-check of the FMS.